

SUMMARY

Senior software engineer, 8+ YOY across high-throughput data infrastructure and developer tooling in Haskell, by applying programming-language theory to production systems at scale.

EMPLOYMENT

Senior Software Engineer, Core **Scrive** **08/2025 — Present**

- Designed and implemented a high-stakes migration, integrating a legacy system serving the main customer base into new AI-based workloads.
- Improved existing CI processes for the main Haskell codebase, reducing integration lead times by ~70%, and increasing deployment velocity.
- Designed and implemented automated methods for internal data traces, easing complex debug scenarios.
- Improved developer experience by facilitating local environment bootstrapping of services.

Software Engineer, Infra **Channable** **02/2021 — 08/2025**

- Refactored infrastructure responsible for importing terabytes of data from external services per day, improving debuggability and observability.
- Designed and implemented AI-assisted categorization using state-of-the-art techniques for mass text classification, improving the existing model performance by 3x.
- Led integration of type-safe user-defined computations into a high-throughput compute pipeline, applying programming-language theory to guarantee safety without sacrificing performance.
- Redesigned core scheduling infrastructure after analyzing its expressivity and usability limits, driving adoption through well-researched proposals and cutting compute overhead by multiple CPU-weeks per day.

Software Engineer **Cargowatch** **02/2018 — 12/2020**

- Implemented a specialized web portal for customer support and designed a DSL for invoicing specification.
- Algorithmically improved the existing automatic invoicing process.

EDUCATION

MSc. BSc. Computing Science **Utrecht University** **2015 — 2020**

- Thesis: Formalized Correctness Proofs of Automatic Differentiation in Coq. Proof using logical relations accompanied by simple language representations and denotational semantics.

PROJECTS

Maintainer, Haskell Language Server <https://github.com/haskell/haskell-language-server>

- Sped up raw indexing speed of GHC-emitted interface files by ~50% through optimizing internal SQLite usage.
- Improved TH handling improving overall memory usage by ~10% and increase editing latency by upto ~90% in some workloads.
- Implemented `lsp-recorder` that can be used to debug realistic editing workloads on arbitrary LSP servers.
- Implemented `hls-export-plugin` for mutating export lists intelligently.

Helium

- Contributed to the Helium Haskell compiler by implementing and integrating missing Haskell2010 features.

PROGRAMMING LANGUAGES AND TECHNOLOGIES

- **Programming Languages:** Haskell, Python, Nix, SQL
- **Technologies:** Git, PostgreSQL, SQLite, Docker
- **Languages:** Dutch, English